Greta Coraglia

https://etagreta.github.io/ coraglia.eta@gmail.com 09/04/1994, Italy

Professional profile

I am a mathematician at the intersection of rigorous research and applied AI development, with specialised expertise in Data Science, Machine Learning, and Generative AI. I am driven by the challenge of turning complex research into practical, responsible solutions that create real social value. A natural communicator and educator, I am passionate about making AI understandable and empowering for everyone. I lead with curiosity, stay calm under pressure, and thrive in collaborative, interdisciplinary environments where connecting people and ideas makes the difference.

Relevant work experience

MIRAI Srl **Mar 2024 - present:** *Co-founder, AI Research Scientist*
Co-founded a start-up with industry experts to help public and private organizations achieve regulatory compliance with transparency and fairness at its core. Led end-to-end R&D across multiple projects, combining applied research with hands-on development to deliver concrete, production-ready solutions. On the Generative AI front, designed and evaluated Retrieval-Augmented Generation (RAG) pipelines, focusing on retrieval quality, answer faithfulness, and end-to-end system performance – translating research findings directly into architectural and implementation improvements. On the Machine Learning side, conducted in-depth assessments of classification algorithms, analysing performance, robustness, and fairness metrics to inform model selection and deployment decisions. Throughout, acted as the bridge between research and engineering, ensuring that theoretical insights were systematically turned into tested, maintainable, and explainable solutions aligned with regulatory requirements.

University of Pavia, AI Dept. **Mar 2025 - present:** *Adjunct professor in Logics for AI*
Lectured for the Logics for Practical Reasoning and AI course in the joint Bachelor's program in Artificial Intelligence (UniPv–UniMi–UniMiB). This role expanded my expertise in causal graphs and honed my public speaking, communication, and organizational abilities.

Segrate City Council **Sep 2020 - present:** *Majority chief whip*
Co-organized a civic movement and was elected to my city's council on a civic list. Currently serving as chief whip of the largest council group and as vice-president of the council. Responsibilities include coordinating council activities on key policy issues, drafting and advancing legislative proposals, negotiating with institutional offices and stakeholders, and organizing public events to inform and actively engage citizens in shaping and implementing public policy.

University of Milan, Philosophy Dept. **Mar 2023 - July 2025:** *Postdoctoral researcher*
Conducted research in the BRIO (Bias, Risk, and Opacity in AI) project, advancing formal methods to assess AI fairness and risk. I co-developed open-source tools in partnership with private companies for real-world applications, such as to credit scoring. I benefitted from working within a multidisciplinary team, combining strong technical problem-solving with effective cross-sector communication.

University of Genoa, Mathematics Dept. **Nov 2019 - Aug 2023:** *Doctoral candidate in Mathematics*
Worked for a Ph.D. in Mathematics with specialization in logic, category theory, type theory, and theoretical computer science. I developed a strong foundation in rigorous problem-solving and abstract

reasoning, with practical applications to formal system design, programming language theory, and algorithmic thinking.

During this time, I also taught a Linear Algebra (Geometry) course to first-year Electronic Engineering students. That strengthened my ability to explain complex technical concepts clearly, while reinforcing my expertise in linear algebra with emphasis on its engineering applications in data modeling.

Technical tools

Python, C, Git, SQL, tools for formal methods (SAT, SMT, Prolog, NuSMV).

Education

University of Genoa *Nov 2019 – Aug 2023*
Ph.D. in Mathematics
Thesis: "Categorical structures for deduction", supervisor: Prof. G. Rosolini
Expertise: logic, category theory, theoretical computer science, type theory

University of Milan *2017 – 2019*
M.Sc. in Mathematics (with honors)
Thesis: "A categorical perspective on Heyting-valued sets", sups: Profs. S. Ghilardi and G. Rosolini
Expertise: logic, category theory, fuzzy logic, automated reasoning, formal methods

University of Milan *2013 – 2017*
B.Sc. in Mathematics
Expertise: algebra, linear algebra, geometry, logic, foundations of mathematics

Awards, certifications, and such

Academic. Won the prize for Best Master's Thesis in Logic (2019) awarded by the Italian Logic Association (AILA). Won several research and travel grants: the Adjoint School participation (2022) in Glasgow, a stay at CIRM Marseille (2022) for "Logic and Higher Structures" and "Linear Logic Winter School", travel to the Symposium on Compositional Structures (SYCO, 2019) in Leicester, and invitations to visit the Mathematics Department at the University of Aberdeen (May 2023), the Laboratoire de Mathématiques de l'Université Savoie Mont Blanc (Nov-Dec 2023), the Compositional Systems and Methods group at Tallinn's University of Technology (May 2022).

Linguistic. British Council Aptis ESOL attested (January 2025) CEFR grade: C2.

Reasoning and numerical. Successfully passed (June 2024) selection tests from the European Personnel Selection office highest grade: EPSO/CAST/P/17/2017 - Information and communication technology (FG IV).

Social involvement

Popularization. I am a strong believer in the necessity of democratizing knowledge: during COVID with some friends we organized community seminars on YouTube, and I gave one on the concept of infinity; in 2020 with a colleague we participated in a high school programme teaching logic (syllogisms and Turing machines) to high-school students; with ItaCa, I helped organizing a YouTube course in category theory with specialists in the topic; I wrote about constructive mathematics in a dissemination journal; with some friends and colleagues we are organizing a public conference to demystify the narrative on AI, to acknowledge its benefits, and to encourage action.

Translating etc (English). Translated two books and contributed to a community initiative preparing Italian subtitles for international works. Provided interpreting support for authors at an international book festival, facilitating communication across languages and cultures.

Last updated: March 2026